MODULE 31 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 1 of 4

Twenty new questions, weighted like the real exam, including four multiple-response questions.

QUESTIONS 20 PASS MARK 15 of 20 HOLD 9 seconds

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How to use this paper

- Pause after each question and commit to an answer before the solution appears.
- Write down your letters. Multiple-response questions need every correct letter.
- Score yourself by **domain** at the end, not only in total.
- For each miss, revisit the module named on the scoring slide.

> ON THE EXAM
>
> The answer teaches the fact. The module teaches the discriminator you missed.

> A Maven project builds in CodeBuild a few times a day. Every build spends eight minutes downloading the same dependencies.
>
> Builds are infrequent and land on different build hosts. The team wants faster builds with the least effort.

What should the DevOps engineer configure?

A Local custom cache for the Maven repository directory

B An S3 cache for the Maven repository directory

C A larger compute type

D Publish the dependencies to CodeArtifact

◆ PAUSE AND DECIDE

What should the DevOps engineer configure?

A Local custom cache for the Maven repository directory

Local caches are best effort and host-bound. Infrequent builds on different hosts rarely hit them.

B An S3 cache for the Maven repository directory

Shared by every build host and persistent between infrequent builds. The right cache for this pattern.

C A larger compute type

The time is spent downloading, not computing.

D Publish the dependencies to CodeArtifact

Still downloads every dependency on every build, just from a different place.

REVISIT MODULE 3

Infrequent builds on different hosts: S3 cache. Frequent builds on warm hosts: local cache.

A pipeline in a tooling account deploys to EC2 instances in a production account with CodeDeploy. Artifacts are encrypted with a customer managed KMS key.

The artifact bucket policy already allows the production account to read. Instances fail to download the revision.

Which steps will fix the deployment? (Choose TWO.)

A. Allow the production instance role to decrypt in the key's policy

B. Grant the instance profile s3:GetObject on the bucket and kms:Decrypt on the key

C. Grant the CodeDeploy service role s3:GetObject

D. Open outbound port 443 in the instances' security group to 0.0.0.0/0

E. Switch the artifact bucket to the AWS managed key for S3

◆ PAUSE AND DECIDE

Which steps will fix the deployment? (Choose TWO.)

A Allow the production instance role to decrypt in the key's policy

The key is in the tooling account, so its policy must admit the production role.

B Grant the instance profile s3:GetObject on the bucket and kms:Decrypt on the key

Cross-account needs both sides: the role's own policy must allow the calls too.

C Grant the CodeDeploy service role s3:GetObject

The agent downloads the bundle using the instance profile, not the service role.

D Open outbound port 443 in the instances' security group to 0.0.0.0/0

The error is access denied, not a network timeout.

E Switch the artifact bucket to the AWS managed key for S3

An AWS managed key can never be used by another account.

REVISIT MODULE 8

Bucket already allows it: the key policy and the caller's own policy are the two locks left.

A V2 pipeline builds and releases a library from GitHub through CodeConnections. It currently runs on every push to main.

The team wants releases only when a version tag such as v2.3.0 is pushed, with no custom code.

What should the DevOps engineer do?

A Add a push trigger with a tag filter that includes v*

B Switch the source to polling every five minutes

C Create an EventBridge rule on CodeCommit tag events

D Use a Lambda function behind a GitHub webhook to start the pipeline on tags

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Add a push trigger with a tag filter that includes v*

V2 triggers filter Git tags natively. No code.

B Switch the source to polling every five minutes

Polling still runs on every change and cannot filter tags.

C Create an EventBridge rule on CodeCommit tag events

The repository is on GitHub, not CodeCommit.

D Use a Lambda function behind a GitHub webhook to start the pipeline on tags

Works, but it is the custom code the team ruled out.

REVISIT MODULE 2

Filter on branches, tags or paths, from a connection source: V2 triggers.

> An Elastic Beanstalk application must send 10% of traffic to each new version for 15 minutes, then shift the rest if healthy.
>
> It must stay in the same environment and roll back automatically if the new version is unhealthy.

Which deployment policy meets these requirements?

A Rolling

B Immutable

C Traffic splitting, 10% for 15 minutes

D Blue/green by swapping environment URLs

◆ PAUSE AND DECIDE

Which deployment policy meets these requirements?

A **Rolling**

Updates batches of existing instances. No traffic percentage or evaluation period.

B **Immutable**

A new group takes traffic without a 10% canary phase.

C **Traffic splitting, 10% for 15 minutes**

A canary share for an evaluation period, then the rest, with automatic rollback.

D **Blue/green by swapping environment URLs**

Uses two environments and moves all traffic at once.

REVISIT MODULE 7

A percentage for a period inside one Beanstalk environment: traffic splitting.

> Developers want to see which unit tests fail, and how the failure rate changes over time, without reading raw build logs.
>
> Tests already produce JUnit XML files during the CodeBuild build phase.

What should the DevOps engineer do with the LEAST effort?

A Upload the XML files to S3 and build a QuickSight dashboard

B Parse the build logs with Logs Insights

C Run the tests in AWS Device Farm

D Add a reports section to the buildspec, sending the XML to a report group

◆ PAUSE AND DECIDE

What should the DevOps engineer do with the LEAST effort?

A Upload the XML files to S3 and build a QuickSight dashboard

A custom reporting pipeline for something CodeBuild does natively.

B Parse the build logs with Logs Insights

The requirement is to avoid reading logs, and parsing is fragile.

C Run the tests in AWS Device Farm

For mobile and browser device testing, not unit tests.

D Add a reports section to the buildspec, sending the XML to a report group

Report groups show each test case and trends over time, from the files already produced.

REVISIT MODULE 3

Test results and trends from CodeBuild: a report group. No custom tooling.

> A stack contains a production RDS database. A future template change could force replacement.
>
> The company must block any update that would replace the database, and keep a snapshot if a replacement is ever forced deliberately.

Which steps meet these requirements? (Choose TWO.)

A Set DeletionPolicy: Delete on the database

B A stack policy denying Update:Replace on the database

C Enable termination protection on the stack

D Set UpdateReplacePolicy: Snapshot on the database

E Set UpdateReplacePolicy: RetainExceptOnCreate

◆ PAUSE AND DECIDE

Which steps meet these requirements? (Choose TWO.)

A Set DeletionPolicy: Delete on the database

The opposite of protection.

B A stack policy denying Update:Replace on the database

Blocks replacement for everyone until deliberately overridden.

C Enable termination protection on the stack

Protects the stack from deletion, not the resource from updates.

D Set UpdateReplacePolicy: Snapshot on the database

If a replacement is forced, the old database is snapshotted.

E Set UpdateReplacePolicy: RetainExceptOnCreate

Not valid for UpdateReplacePolicy; that option exists only for DeletionPolicy.

REVISIT MODULE 9

Block the update: stack policy. Protect data if it happens anyway: UpdateReplacePolicy.

> Every EC2 instance, including instances launched in future, must run the CloudWatch agent with a configuration stored in Parameter Store.
>
> If someone stops the agent, it must be restored automatically.

What should the DevOps engineer use?

A Run Command, once, against all current instances

B State Manager associations that install and configure the agent on a schedule

C User data in every launch template

D Bake the agent into a new AMI

◆ PAUSE AND DECIDE

What should the DevOps engineer use?

A Run Command, once, against all current instances

Misses future instances, and never restores a stopped agent.

B State Manager associations that install and configure the agent on a schedule

Applies to new instances and re-applies continuously, correcting drift.

C User data in every launch template

Runs once at launch; a stopped agent stays stopped.

D Bake the agent into a new AMI

Helps new instances start with it, but nothing restores a stopped agent.

REVISIT MODULE 13

'Including future' and 'restored automatically': State Manager.

> Developers in the Sandbox OU may only launch t3 and t4g instance types. They launch through the console, the CLI and CloudFormation.
>
> The company wants a preventive control across every Sandbox account with the least overhead.

What should the DevOps engineer implement?

A. A CloudFormation Guard Hook checking instance types

B. An AWS Config rule for approved instance types

C. An SCP on the Sandbox OU denying ec2:RunInstances unless ec2:InstanceType matches t3.* or t4g.*

D. A Service Catalog portfolio of approved instances

◆ PAUSE AND DECIDE

What should the DevOps engineer implement?

A A CloudFormation Guard Hook checking instance types

Only covers CloudFormation. Console and CLI launches bypass it.

B An AWS Config rule for approved instance types

Detective: the instance already exists when it is flagged.

C An SCP on the Sandbox OU denying ec2:RunInstances unless ec2:InstanceType matches t3.* or t4g.*

Applies to every API call, however made, in every account in the OU.

D A Service Catalog portfolio of approved instances

Offers approved products but does not stop direct launches.

REVISIT MODULE 12

'Console, CLI and CloudFormation' rules out a Hook. Every API call, preventive: an SCP.

> A legacy application runs on EC2 with its data on EBS volumes and cannot be re-architected.
>
> After a Regional failure it must lose no more than one minute of data and be running within four hours, at the lowest cost.

Which solution meets these requirements?

A AWS Elastic Disaster Recovery replicating to a second Region

B Hourly AWS Backup copies to a second Region

C A full-size warm standby fleet in a second Region

D Multi-Region active-active

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A AWS Elastic Disaster Recovery replicating to a second Region

Seconds of RPO with a cheap staging area; recovery well within four hours.

B Hourly AWS Backup copies to a second Region

Up to an hour of data lost. Fails the one-minute RPO.

C A full-size warm standby fleet in a second Region

Meets the targets, at far higher cost than needed.

D Multi-Region active-active

The most expensive option, and needs re-architecture.

REVISIT MODULE 16

The one-minute RPO rules out backups; the four-hour RTO rules out paying for running copies.

> A web tier on an Auto Scaling group behind an ALB serves report requests lasting up to two minutes.
>
> During scale-in, users see 502 errors when instances are removed mid-request.

What should the DevOps engineer change?

A Add a launch lifecycle hook

B Turn off scale-in

C Use larger instances

D Set the target group's deregistration delay above two minutes

◆ PAUSE AND DECIDE

What should the DevOps engineer change?

A Add a launch lifecycle hook

Acts at launch, not while an instance is being removed.

B Turn off scale-in

Stops the errors by giving up scaling, and the cost savings with it.

C Use larger instances

Requests are cut off on removal whatever the instance size.

D Set the target group's deregistration delay above two minutes

In-flight requests finish before the instance is terminated.

REVISIT MODULE 14

Requests cut off on removal: deregistration delay longer than the longest request.

A web application runs active-passive in two Regions. Static assets are in S3 and user data in DynamoDB.

If the primary Region fails, static content must fail over automatically and user data must already be in the secondary Region.

Which steps meet these requirements? (Choose TWO.)

A S3 replication, with a CloudFront origin group across both buckets

B A nightly job copying S3 objects to the second Region

C A DynamoDB global table across both Regions

D Daily DynamoDB backups copied to the second Region

E Route 53 simple routing to the primary Region

◆ PAUSE AND DECIDE

Which steps meet these requirements? (Choose TWO.)

A S3 replication, with a CloudFront origin group across both buckets

Content is copied, and CloudFront retries the secondary bucket automatically.

B A nightly job copying S3 objects to the second Region

Up to a day stale, and nothing fails over automatically.

C A DynamoDB global table across both Regions

User data is already present in the secondary Region.

D Daily DynamoDB backups copied to the second Region

Data must be restored first, and up to a day is lost.

E Route 53 simple routing to the primary Region

Simple routing has no failover.

REVISIT MODULE 14

Two halves: static content fails over at the edge; data is already replicated.

An application already writes a log line containing LOGIN_FAILED for each failed sign-in to CloudWatch Logs.

Security wants an alarm when failed sign-ins exceed 50 in five minutes. The code cannot change.

What should the DevOps engineer do?

A. A metric filter on the log group, and an alarm on the metric

B. Rewrite the logging to use the embedded metric format

C. A subscription filter to a Lambda function that counts failures

D. A Logs Insights query run by hand every five minutes

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A A metric filter on the log group, and an alarm on the metric

Turns existing log lines into a metric, with no code change.

B Rewrite the logging to use the embedded metric format

A code change, which is ruled out.

C A subscription filter to a Lambda function that counts failures

Custom code for what a metric filter does natively.

D A Logs Insights query run by hand every five minutes

Manual, and not an alarm.

REVISIT MODULE 17

Value already in the logs, no code change: metric filter.

> A service pages on-call when CPU is high, and again when latency is high. Most pages are false alarms from one or the other alone.
>
> The team wants a page only when high CPU and high latency happen together.

What should the DevOps engineer configure?

A Send both alarms to the same SNS topic

B An anomaly detection alarm on CPU

C A composite alarm with an AND rule, carrying the only notification

D A metric math alarm on CPU plus latency

◆ PAUSE AND DECIDE

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What should the DevOps engineer configure?

A Send both alarms to the same SNS topic

Still pages for each alarm alone.

B An anomaly detection alarm on CPU

Changes one threshold; still pages on CPU alone.

C A composite alarm with an AND rule, carrying the only notification

Fires only when both child alarms are in ALARM.

D A metric math alarm on CPU plus latency

Adding unrelated units produces a meaningless number.

REVISIT MODULE 18

Page only when conditions coincide: composite alarm with AND.

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> A team runs many services on ECS with Fargate. It needs CPU, memory and network metrics per service and per task, with ready-made dashboards.
>
> It wants the least operational effort.

What should the DevOps engineer enable?

A. A CloudWatch agent sidecar in every task definition

B. CloudWatch Container Insights for the cluster

C. AWS X-Ray tracing

D. A self-managed Prometheus server

◆ PAUSE AND DECIDE

What should the DevOps engineer enable?

A. A CloudWatch agent sidecar in every task definition

Changes every task definition to rebuild what is built in.

B. CloudWatch Container Insights for the cluster

Per-service and per-task metrics with automatic dashboards.

C. AWS X-Ray tracing

Traces requests; does not report resource metrics.

D. A self-managed Prometheus server

A server to run, for metrics available natively.

REVISIT MODULE 17

Container resource metrics with dashboards, least effort: Container Insights.

> The team wants an alert when any EC2 instance tagged env=prod is stopped.
>
> An EventBridge rule on EC2 state-change events with a pattern on the tag never matches.

Why, and what is the fix?

A The pattern is case-sensitive; fix the case of the tag key

B State-change events are only sent for Auto Scaling instances

C The rule needs a trail to receive state-change events

D The event does not carry tags; match state 'stopped' and have a Lambda target check the tag

◆ PAUSE AND DECIDE

Why, and what is the fix?

A The pattern is case-sensitive; fix the case of the tag key

Case is not the problem. The tag is not in the event at all.

B State-change events are only sent for Auto Scaling instances

Every instance emits them.

C The rule needs a trail to receive state-change events

State changes come from EC2 directly; a trail is for API-call events.

D The event does not carry tags; match state 'stopped' and have a Lambda target check the tag

Enrich the event: look up the tag, then notify only for production.

REVISIT MODULE 21

Events carry what the service puts in them. Instance tags are not in state-change events.

> All new EBS volumes in every account must be encrypted, without changing any team's templates or scripts.
>
> Existing unencrypted volumes must be reported for follow-up.

Which steps meet these requirements? (Choose TWO.)

A An SCP denying ec2:CreateVolume unless encryption is requested

B Amazon Macie in every account

C Turn on EBS encryption by default in every account and Region

D Amazon GuardDuty in every account

E An AWS Config rule reporting unencrypted volumes, through a conformance pack

◆ PAUSE AND DECIDE

Which steps meet these requirements? (Choose TWO.)

A An SCP denying ec2:CreateVolume unless encryption is requested

Breaks every template that does not request encryption: the thing ruled out.

B Amazon Macie in every account

Macie examines S3 objects, not EBS volumes.

C Turn on EBS encryption by default in every account and Region

New volumes are encrypted automatically, with no template changes.

D Amazon GuardDuty in every account

Detects threats, not unencrypted volumes.

E An AWS Config rule reporting unencrypted volumes, through a conformance pack

Reports existing unencrypted volumes across accounts.

REVISIT MODULE 22

'Without changing templates' rules out a deny. Encryption by default fixes new; Config reports old.

A CodeBuild project was moved into private subnets with no NAT gateway, to reach an internal database.

Builds now fail at start-up with a timeout fetching a Secrets Manager secret. The service role's permissions are unchanged and correct.

What should the DevOps engineer do?

A Enable privileged mode

B Add kms:Decrypt to the service role

C Add a Secrets Manager interface endpoint with private DNS to the VPC

D Add an S3 gateway endpoint

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Enable privileged mode

For Docker builds. Unrelated.

B Add kms:Decrypt to the service role

A timeout, not access denied; permissions are correct.

C Add a Secrets Manager interface endpoint with private DNS to the VPC

Gives the build a private path to Secrets Manager.

D Add an S3 gateway endpoint

Secrets Manager is not reached through S3.

REVISIT MODULE 23

Timeout after moving into a VPC, permissions fine: missing network path. Add the service's endpoint.

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> A GitHub Actions workflow deploys to AWS using an IAM user's access keys stored as GitHub secrets.
>
> Security wants no long-lived keys, and only the repository's main branch able to deploy.

What should the DevOps engineer do?

A. An IAM OIDC provider for GitHub, and a role whose trust allows only this repository's main branch

B. Rotate the access keys every 30 days

C. IAM Roles Anywhere with a certificate stored in GitHub

D. An IAM Identity Center user for the workflow

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A An IAM OIDC provider for GitHub, and a role whose trust allows only this repository's main branch

Short-lived credentials per run, scoped by the token's subject claim.

B Rotate the access keys every 30 days

Still long-lived keys, just younger.

C IAM Roles Anywhere with a certificate stored in GitHub

For on-premises workloads; storing a certificate is another long-lived secret.

D An IAM Identity Center user for the workflow

For people, not CI systems.

REVISIT MODULE 24

CI system, no keys: OIDC federation. The subject condition limits it to one repo and branch.

> EBS snapshots encrypted with the AWS managed aws/ebs key must be shared with a disaster recovery account.
>
> Sharing them fails.

What should the DevOps engineer do?

A Share the snapshots through AWS RAM

B Make the snapshots public

C Add the DR account to the aws/ebs key policy

D Copy each snapshot re-encrypted with a customer managed key shared with the DR account, then share the copy

◆ PAUSE AND DECIDE

What should the DevOps engineer do?

A Share the snapshots through AWS RAM

The key is still the AWS managed one, which the other account cannot use.

B Make the snapshots public

Never, and encrypted snapshots cannot be public anyway.

C Add the DR account to the aws/ebs key policy

AWS managed key policies cannot be edited.

D Copy each snapshot re-encrypted with a customer managed key shared with the DR account, then share the copy

The copy uses a key the DR account is allowed to use.

REVISIT MODULE 25

Cross-account plus AWS managed key: re-encrypt with a customer managed key.

> A login endpoint behind CloudFront faces credential stuffing from many addresses, and the business serves only three countries.
>
> The team wants to block other countries and slow any single address hammering /login.

What should the DevOps engineer configure?

A Network ACL rules for each attacking address

B A WAF web ACL with a geographic match rule and a rate-based rule scoped to /login

C Security group rules on the origin

D AWS Network Firewall in front of the origin

◆ PAUSE AND DECIDE

What should the DevOps engineer configure?

A Network ACL rules for each attacking address

Endless manual lists, and blind to countries and paths.

B A WAF web ACL with a geographic match rule and a rate-based rule scoped to /login

Blocks by country, and limits each address on the login path only.

C Security group rules on the origin

Addresses and ports only; no countries or paths.

D AWS Network Firewall in front of the origin

A network-layer inspector; this is an HTTP problem at the edge.

REVISIT MODULE 26

Countries and request paths are HTTP-level ideas: WAF.

Answer key

- **Answer key:** 1 B, 2 AB, 3 A, 4 C, 5 D, 6 BD, 7 B, 8 C, 9 A, 10 D
- **Answer key:** 11 AC, 12 A, 13 C, 14 B, 15 D, 16 CE, 17 C, 18 A, 19 D, 20 B
- Multiple response (2, 6, 11, 16) score only if both letters are right.
- 15 or more: on track. 13 or 14: close. 12 or fewer: revisit weak domains first.

> ON THE EXAM
>
> Each solution slide named the module to revisit. Use it for every miss.

Scoring by domain

DOMAIN	QUESTIONS	MODULES TO REVISIT
D1 SDLC Automation	1, 2, 3, 4, 5	2, 3, 7, 8
D2 Configuration Management and IaC	6, 7, 8	9, 12, 13
D3 Resilient Cloud Solutions	9, 10, 11	14, 16
D4 Monitoring and Logging	12, 13, 14	17, 18
D5 Incident and Event Response	15, 16, 17	21, 22, 23
D6 Security and Compliance	18, 19, 20	24, 25, 26

ON THE EXAM

Two misses in one domain matter more than three spread across six.